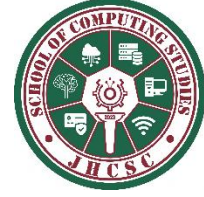




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Smart Student Panic Button and Emergency Response System with Campus Incident Mapping and SMS Backup Notification

A CAPSTONE PROJECT
Presented to the Faculty of
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CHAPTER I

THE PROBLEM AND ITS SCOPE

Introduction

The safety and security of students, faculty, staff and visitors is one of the paramount responsibilities of educational institutions. Emergencies that require an immediate response on campus include accidents, medical emergencies or security threats. But traditional reporting methods, such as phone calls and face-to-face communication, may result in delays and affect the timely delivery of emergency assistance. With developments in Information Technology, the educational institutions are adopting digital solutions that allow for faster communication, real-time monitoring and more efficient emergency management.

Through technological advancement such as the Internet of things (IoT), smart campus technology has been adopted that utilize connected devices, digital services and communications networks to support effective emergency response. Emergency services are able to communicate more efficiently, respond quicker and have real time awareness on any incidents with the adoption of smart campus technology. IoT based emergency management systems increase the awareness about emergencies and hasten response time through a network of linked devices and web-based tracking systems as emphasized by Delnevo et al. (2024). Additionally, Beltran et al. (2020), state that wireless micro-controller-based communications for emergency alert devices are efficient and low-cost system that are provided for the warning process during a given emergency situation.

At the Main Campus of J.H. Cerilles State College (JHCSC), emergency incidents are mostly reported through telephone calls or personal verbal report to Security Office or Disaster Risk Reduction and Management Office (DRRMO). This manual system has been effective but may cause delayed reporting of incidents and difficulties in locating exact spot of the incidents during emergencies. To resolve these problems, this study developed a system called as smart student panic button and emergency response system with campus incident mapping and SMS backup notification. The system comprises of ESP32 based panic button gadgets, web-based monitoring platform, campus incident mapping and SMS based on GSM as a backup notification in reporting and locating incident in JHCSC Main Campus.

Project Context

The Josefina H. Cerilles State College (JHCSC) Main Campus has experienced several emergency situations, such as accidents, medical emergencies and security-related incidents that require immediate attention to ensure the safety and welfare of the students, faculty, staff and visitors. However, most of the reported emergency situations are through calls to or conversations with the security guard/security office and directly with the DRRMO personnel. Even though a phone call or direct verbal communication is an effective means of relaying emergency-related news, the practice remains manual, which may delay prompt reporting and response. Sometimes, the reporter may not give full or accurate information, and responders may find it hard to determine where an incident is happening. This practice wastes precious time trying to find the right people or channel of communication.

This study proposes the development of the Smart Student Panic Button and Emergency Response System with Campus Incident Mapping and SMS Backup Notification for JHCSC Main Campus in response to these challenges. The proposed system is to install smart panic button devices at various points on the campus. The relevant emergency button allows users to report emergencies instantly. Each device is controlled by a main controller which is an ESP32 Development Board. The button inputs are processed by the controller, and then the emergency alerts are sent to a web-based monitoring platform using a wireless network. The device is powered by a 12V DC Power Adapter and uses an LM2596 Buck Converter (Voltage Regulator) to provide the stable operating voltage needed by the electronic components. The system is equipped with a SIM800L GSM Module to ensure constant emergency communication in case of internet failure by automatically sending SMS messages to pre-selected responders as a backup communication method.

The proposed system is based on the integration of embedded systems, Internet of Things (IoT) technology, campus incident mapping and backup GSM communication into a single web-based monitoring system to improve emergency reporting by providing real-time incident notification and precise location identification. This will allow the DRRMO and other emergency responders to get the data as timely as possible, coordinate the efforts of responders effectively and efficiently, and provide assistance at the earliest possible time

and opportunity, thereby further enhancing the safety and emergency preparedness of the JHCSC Main campus.

Purpose and Description

The Smart Student Panic Button and Emergency Response System with Campus Incident Mapping and SMS Backup Notification intends to offer a faster, more efficient and real-time emergency reporting and response system within the J.H. Cerilles State College (JHCSC) Main Campus than the current manual emergency notification process. The system is designed to cut down on time in emergency situations by enabling instant communication between campus users and the appropriate emergency responders using strategically deployed smart panic button devices at the STE, SAFES, SAS Department, Admin Building, Computer Laboratory, Gymnasium, Engineering Building and Library.

The proposed system consists of ESP32 based panic button devices, web-based monitoring platform, campus incident mapping and a GSM based SMS backup notification mechanism. Each panic button device has three dedicated emergency buttons for Critical Emergency, Medical Emergency, and Public Safety Emergency, enabling users to report the appropriate type of emergency immediately. Upon pressing a button, the ESP32 receives the selected emergency category and transmits the alert to the web-based monitoring platform, where the incident appears on the campus map and is recorded in real-time for monitoring by the DRRMO/Security Office. In case of a loss of internet connectivity, the GSM module automatically sends an SMS backup notification to ensure that emergency alerts are still delivered promptly. The proposed system enhances the emergency coordination and offers an expedited emergency response in JHCSC Main Campus by means of emergency categorization, real-time monitoring, incident mapping and back-up communication.

Objectives of the Study

General Objective

To develop a Smart Student Panic Button and Emergency Response System with Campus Incident Mapping and SMS Backup Notification that will enable faster emergency reporting, emergency categorization, incident location identification and coordinated emergency response within the J.H. Cerilles State College (JHCSC) Main Campus.

Specific Objectives

This project specifically aims to:

1. To design an ESP32-based panic button device with three dedicated emergency buttons for Critical Emergency, Medical Emergency, and Public Safety Emergency to transmit emergency alert signals from designated areas of the campus.
2. To develop a web application monitoring interface in which DRRMO will receive alerts, be able to see incident locations on the campus incident map, and control and monitor emergency data of incidents.
3. To integrate a GSM-based backup notification system by which the DRRMO can receive the emergency alert through text messages when the internet is not available.
4. To implement a campus incident mapping feature that indicates the location of triggered panic button devices to speed up the emergency response.
5. To test and evaluate the functionality and effectiveness of the planned system with respect to sending emergency alerts, GSM-based backup notifications, campus incident maps, and web applications.

Scope and Limitations

The Smart Student Panic Button and Emergency Response System with Campus Incident Mapping and SMS Backup Notification is an IoT-based emergency reporting system for the J.H. Cerilles State College (JHCSC) Main Campus. The system uses ESP32 based devices as panic button devices placed at specific places on the campus such as STE, SAFES, SAS Department, Admin Building, Computer Laboratory, Gymnasium, Engineering Building and Library. Each device has three dedicated emergency buttons for Critical Emergency, Medical Emergency and Public Safety Emergency. When the system is activated, the selected emergency alert will be transmitted to an online monitoring platform available to the DRRMO that displays the incident location in campus incident mapping, records the emergency report and sends GSM based SMS back up notifications in case of unavailability of internet connectivity. The School Clinic is notified of medical related emergencies to provide appropriate medical response.

The system is activated manually only by the installed panic button devices; it does not automatically detect emergency situations. The location of an incident is based on the

pre-registered location of the panic button device, and not based on GPS technology. The operation of the system is dependent on the availability of electrical power, internet connectivity for real time communication and GSM cellular network coverage for SMS backup notifications. The proposed system is also intended to be used only at the JHCSC Main Campus and does not interface with external emergency response agencies or automatic dispatch services.

Definition of Terms

ESP32. a microcontroller board which serves as the primary processing unit of the panic button device. It receives inputs from the button, processing the emergency requests, establishing connections to the network, and controlling the outputs of the device.

Internet of Things (IoT). used for the panic button device and monitoring platform to interact in time of emergency.

Smart Student Panic Button and Emergency Response System. is an IoT-based emergency notification system which aim to provide faster alert, identification of location of emergency, and coordination of emergency response within the JHCSC Main Campus.

Panic Button Device. a hardware device which is installed in certain campus locations that will manually activate the alert button in time of needed assistance.

Campus Incident Mapping. an application feature which identifies the location of the incident by getting the assigned location of the activated panic button device and displays it on the web-based monitoring platform.

GSM Module. a communications component that provides SMS-sending capability, useful as a back-up in case there are no available internet connection.

SMS Backup Notification. A communication method for sending out emergency notification via SMS as part of the emergency notification protocol as a backup to provide notification during possible interruption of internet connectivity.

Web-Based Monitoring Platform. an online system accessible to NDRRMO/security office personnel that serves as interface for viewing and monitoring of alerts, incident and their location and records.

NDRRMO/Security Officer. responsible agency in the campus for receiving emergency alerts, monitoring of ongoing emergencies, and coordination of emergency response and operations.

Critical. Emergency. Emergency that poses imminent threat to life or serious injury and requires urgent response by the NDRRMO and the School Clinic.

Medical Emergency. An illness or injury emergency that requires medical assistance from the School Clinic but is not life threatening at the present time.

Public Safety Emergency. An emergency involving security, fire, accidents or other public safety concerns that requires primarily a response by the NDRRMO/Security Office;

Emergency Alert. a notification sent out by the system upon receiving a press from the panic button.

CHAPTER II

REVIEW OF RELATED STUDIES AND SYSTEMS

In this chapter we explain the software, hardware, programming development environment and other works related to build the Smart Student Panic Button and Emergency Response System with Campus Incident Mapping and SMS Backup Notification system.

Technical Background

With the ongoing development of Internet of Things (IoT) technology, interconnected devices are able to transmit and receive information in real-time via the Internet. Internet of Things (IoT) is widely used in all industries, including the education sector in which it serves to develop smart campus solutions for improved operational efficiency, safety and incident management. This can be achieved through smart campus technologies comprising connected devices, digital platforms, and communication system in real-time to ensure improved situation awareness and quick response to the reported emergencies (Delnevo et al., 2024).

An embedded system provides the basic processor and controls which embedded with an IOT based application device; An Embedded System combines computer hardware and software elements. An Embedded system inputs, stores data, and generates the appropriate outputs. An ESP32 microcontroller is employed to be the main core of this panic button device as it is built with wireless communications capabilities, powerful processing capabilities, and IoT enabled application is well suited. This proposed system features three distinct emergency buttons to trigger Critical Emergency, Medical Emergency, and Public Safety Emergency to be monitored and acted upon appropriately. Based on previous research, the application of a microcontroller for an emergency notification device is one of the affordable and efficient ways (Beltran et al., 2020). Communication is a fundamental factor to support timely action during an emergency.

The integration of IoT devices with web-based monitoring systems enable to transmit, monitor, and handle emergency alert in real-time to ensure efficient emergency communication and rapid response (Delnevo et al., 2024). To overcome this, GSM technology serves as a redundant notification technology to assure communication in absence of internet connection. Through the incorporation of GSM modules into the

microcontroller-based devices, the alerts could be sent out in an SMS to the appropriate persons/ responders in an event where the network is lost (Beltran et al., 2020). A web-based monitoring platform is introduced to centralize viewing of emergency alerts, mapping of emergency incident location via the campus incident mapping and managing emergency records for faster emergency response.

Hardware Specification

Hardware	Description
ESP32 Development Board	The main controller that processes emergency inputs and manages system communication.
Push Button Switch	The input device used to activate the emergency alert.
SIM800L GSM Module	Sends SMS notifications as a backup when internet connection is unavailable.
Buzzer Module	Provides an audible alert when an emergency is triggered.
LM2596 Buck Converter	Regulates voltage supplied to the electronic components.
12V Power Supply Adapter	Provides power for the system components.
Enclosure/Protective Case	Protects the electronic components of the device.
WiFi Network/Router	Provides internet connection for sending data to the monitoring platform.
Computer/Laptop	Used by the DRRMO/security office to access and monitor emergency alerts.

Software Specification

Software	Description
Server Environment	
Operating system	Windows 10 or 11
Local Server	Laragon
Web Server	Apache HTTP Server
Database Management System	MySQL

Backend Language	PHP
Development Framework	None (Native PHP)
Development Environment	
Text Editor/IDE	Visual Studio Code, Arduino IDE
Programming Language	PHP, JavaScript, C/C++
Client Environment	
Device	Computer/Laptop, ESP32-based Panic Button Device
Browser	Google Chrome, Microsoft Edge
Internet	Reliable internet connection for communication between ESP32 device and web-based monitoring platform

Programming Environment

The Smart Student Panic Button and Emergency Response System with Campus Incident Mapping and SMS Backup Notification integrates embedded system and internet development for the design of a web-enabled IoT-based system that deals with emergency reporting and response. The panic button is mainly controlled by an ESP32 microcontroller which is responsible for receiving signals from three different panic buttons, handling wireless data and forwarding the alerts to the remote web system. A SIM800L GSM module has been incorporated for SMS alerts in instances where network service is unavailable. A firmware program written in C++ is deployed on the ESP32 controller. It is compiled and developed using the Arduino IDE, and overseas operations such as the device's functionalities, event trigger handling, the setting up of Wi-Fi communication to a distant server, and initiating the GSM connection. WiFi.h library among other libraries are incorporated in the program, primarily for facilitating wireless communications with the monitoring server. The system's web front-end utilizes Native PHP, HTML, CSS, and JavaScript for dynamic web functionality, while MySQL has been designated as the system database for all records. Laragon has been adopted as the local web server and Visual Studio Code is used as the preferred IDE to run the development and modification of the

web front-end. This web application enables the DRRMO personnel to see the emergency requests that have come in and respond to the incident locations based on the campus incident map view.

Related Studies

Foreign Studies

Recently research on using Internet of Things (IoT) and digital communication technologies has presented benefits in emergency response within educational institutes and the public spheres. The emergency management systems with Internet of Things (IoT) use linked devices with wireless communication to manage emergencies. These systems consist of smart connected devices and the transmission technology allows for constant supervision, faster relay of information and coherent emergency response, Alotaibi et al. (2022), Flores-Salgado et al. (2024), Shu et al. (2023). These researches indicate that having devices integrated in a centralized monitoring station provide with better real time data to enable the situation assessment, expedite the sending of notification of any emergency and also support decision-making processes for any emergency response plan. Corresponding to the result above, Repanovici et al. (2022) suggests that integrated alert systems can make the coordination between responders more effective and minimize reporting time through sending a precise electronic alert to the correct people at the right time.

Moreover, the application of the digital emergency reports application has been extensively used as a communication tool to enhance user and emergency service provider interactions. For instance, the mobile panic button application, which allows the reporting of emergencies through smartphone devices, provides a fast, accurate reporting service as shown by Nasir et al. (2022). Mobile panic button applications are able to alert responders to an emergency more rapidly and accurately because end users can send alerts to emergency personnel instantly through their smart phones. The collective efforts demonstrated that by integrating digital technologies, the IoT devices, and central monitoring stations, communication efficiency and responsiveness for emergencies could be improved. However, the existing works primarily focus on either mobile application or generic IoT infrastructure and do not include a system for providing a campus incident mapping system, emergency type, or using the ESP32-based panic button device and SMS

backup notification using GSM technology at the campus level which are the innovative points of this research.

Local Studies

Local studies also demonstrated the application of Internet of Things technology and the digitalization to better provide services in emergency reporting and public safety management. The IoT-based Barangay Panic Button System by Tello et al. (2026) using an ESP32-C3 microcontroller with e-mail notification validation and cloud logging was able to suggest that microcontroller-based devices help in making emergency devices be effective in recording incidents and reporting. Similarly, the IoT and Mobile technology approach of Ruiz-Ocampo et al. (2026) helps the university campus manage safety through monitoring the incident real-time and having the reported events available online. The suggested application of ESP32 embedded systems and IoT communications by aforementioned local research support the study.

Further local research attempts to provide a system to digitalize the emergency response and monitoring for better communication and coordination among the responding team. In the paper by Lagman et al. (2022) they developed an I-Respond mobile application to digitalize the communication of emergency incident and speed up the response of team. Baltazar et al. (2022) they provided a Disaster and Crime Reporting and Monitoring system with decision support. Their system aims to centralize monitoring of incidents and speed up emergency response. Similarly, the work of Montefalcon et al. (2021) developed the IRespondPH, a mobile and web-based application to facilitate and improve the communication, information management, and emergency response in the event of disaster. These studies, all show how digital tools aid communication and managing of emergency incident, yet local studies mainly focus on developing a mobile and web-based applications and it does not feature the ESP32-based panic button device, campus incident mapping, 3-tier emergency classification and GSM-based SMS backup notification of the proposed Smart Student Panic Button and Emergency Response System.

Synthesis

According to the reviewed research, combining IoT technologies, communication systems and digital platforms can reduce response time and facilitate better emergency management and response, incident monitoring and reporting. Earlier researches like that

on mobile emergency warning applications, panic button systems and safety solutions using IoT devices have already proven that an automatic alerting system can speed up emergency response compared to manual emergency communication system; it can quick the response time through providing real-time access to the event and reduce the potential delays in providing information to the relevant responders. They highlighted the importance of real-time information communication, and centralization of event information and monitoring platform as a way of efficient emergency response.

Yet currently, the vast majority of existing studies are concentrated on mobile application-oriented solutions or generalize disaster surveillance, or generalize safety administration solutions. For a few systems that support issuing emergency alerts, they did not consider the existence of a specific hardware panic button for the campus scene. Furthermore, only a few systems also consider incorporating the mapping feature for campus incident reporting and provide backup solutions that are compatible for scenarios where internet access is disabled.

This research seeks to develop an innovative solution to mitigate such issues, namely, the Smart Student Panic Button and Emergency Response System with Campus Incident Mapping and SMS Backup Notification for JHCSC Main Campus. The envisioned solution shall integrate an ESP32-based panic button device, a web-based interface for monitoring incident reports and for campus location tracking, and a GSM SMS service as a backup notification mechanism for the system to allow efficient, and quick response and access for students during emergencies in JHCSC.

CHAPTER III

DESIGN AND METHODOLOGY

This chapter includes the requirements analysis, methodology used, and the system design diagrams, including 4D methodology, current technical situation, context diagram, conceptual framework, activity diagram, use case diagram, entity relationship diagram, structured diagram, and system architecture of the study.

Requirements Analysis

The development of the Smart Student Panic Button and Emergency Response System utilizes the 4D methodology where the first stage involves Requirements Analysis under Define stage to provide a system to provide a platform to enhance the campus emergency response by enable the swift reporting of events, the monitoring of those events in real-time, and the swift coordination among the emergency responders.

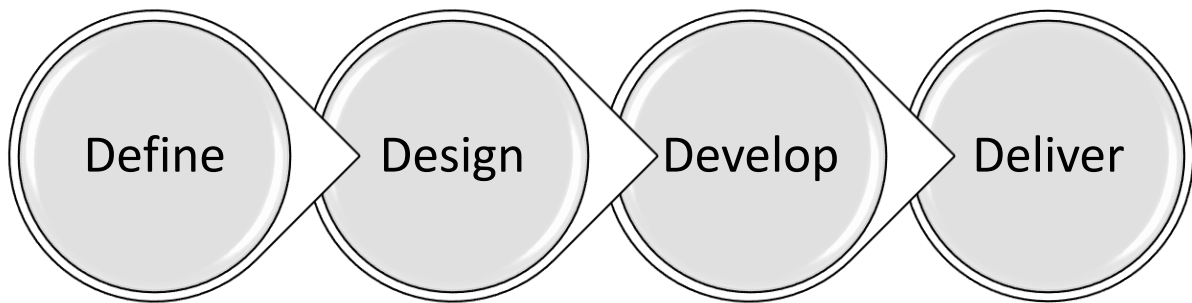


Figure 1. 4D Methodology

The 4D model presented in figure 1. is the methodology which is adopted by the proponents in the creation of the Smart Student Panic Button and Emergency Response System. The methodology can be used by proponents as a map indicating the steps that should be taken to develop and successfully deliver the system. The model represents the 4 phases to which the proponents are advised when it comes to delivering this comprehensive smart system.

Define Phase

The study started with an interview conducted by the proponents, gather the necessary data on the J. H. Cerilles State College-Disaster Risk Reduction Management Office's current operations.



Image 1. Interview with the DRRMO Staff

The proponents conducted an interview with Mr. Greg Lorenze Dominguez Quiling, DRRMO Staff. He explained that campus emergency reporting still relies on phone calls and manual reporting, causing delays and inefficient incident record management. Based on these findings, the proponents identified the need for an effective emergency response system that enables real-time incident reporting and organized data management. The interview served as the primary basis for defining the system requirements of the Smart Student Panic Button and Emergency Response System with SMS Backup for Campus Emergency Management.



Image 2. Interview with the Nurse Assistant

The proponents interviewed Ms. Anzeille Mae E. Patigayon, Nurse Assistant. She explained that the school clinic relies on manual reports from the DRRMO or campus staff before responding to emergencies, which may cause delays in preparing medical personnel and equipment. These findings highlighted the need for a system that provides real-time emergency alerts to improve the clinic's response. The interview served as a basis for developing the Smart Student Panic Button and Emergency Response System with SMS Backup for Campus Emergency Management.

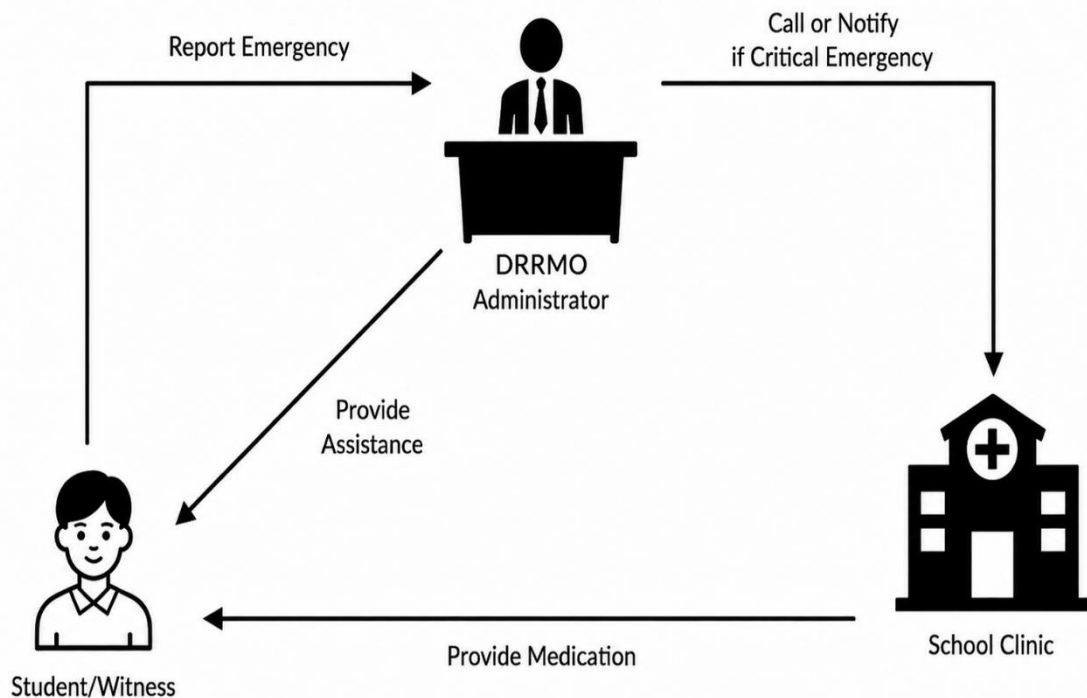


Figure 2. Current Technical Situation

Current Existing System for Emergency Reporting. Figure 2 below presents the present existing system of reporting in times of emergency at the Josefina H. Cerilles State College (JHCSC) Main Campus. The system has a student, who reports in to the DRRMO Administrator, the administrator responds to the report accordingly. In times of Critical Emergencies, the DRRMO Administrator contacts the School Clinic for medical assistance. Given that all communications made in reporting as well as response is all done manually there might be possible delays in the report and response to give assistance in emergency cases. There are no proper, integrated digital facility for monitoring, managing, keeping records, for emergency report in the existing process.

Design Phase

This phase is the second stage of the 4D model, in which the system's structure and overall concept are illustrated. It includes diagrams and explanations that present the system's overall features and functions.

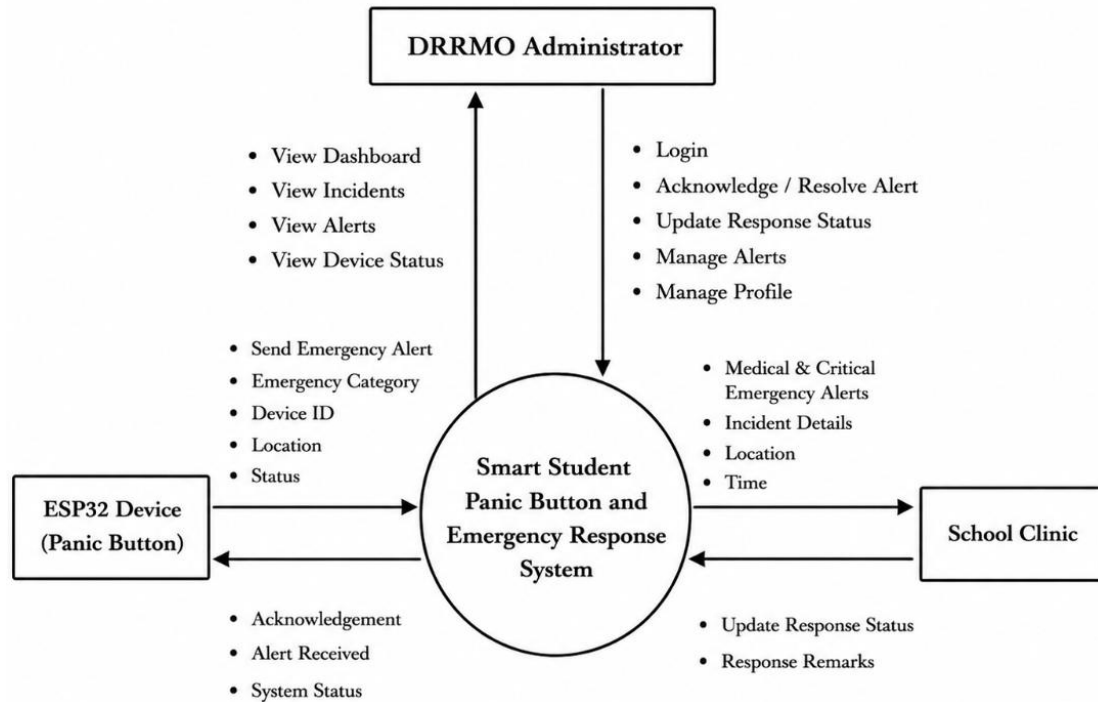


Figure 3. Context Diagram

Figure 3 illustrates the interaction among the DRRMO Administrator, ESP32 Device (Panic Button), School Clinic and the Smart Student Panic Button and Emergency Response System. The DRRMO Administrator logs into the system to receive and monitor emergency alerts, acknowledge or resolve incidents, update response status, manage alerts and user profiles and view the dashboard, incident reports, alerts and device status. The main input device for sending the emergency alert is the ESP32 Device, which sends the selected emergency category, device ID, location and device status each time one of the three dedicated emergency buttons (Critical Emergency, Medical Emergency or Public Safety Emergency) is pressed and acknowledgement, alert confirmation and system status is received from the monitoring platform. In case of medical and critical emergencies, the emergency alert, incident details, location and time of occurrence is received from the system by the School Clinic and the response status and response remarks are updated after providing the necessary medical assistance. Through these interactions, the proposed system offers centralized emergency monitoring, real time communication and better coordination between the DRRMO and the School Clinic for a faster and more efficient emergency response within the J.H. Cerilles State College (JHCSC) Main Campus.

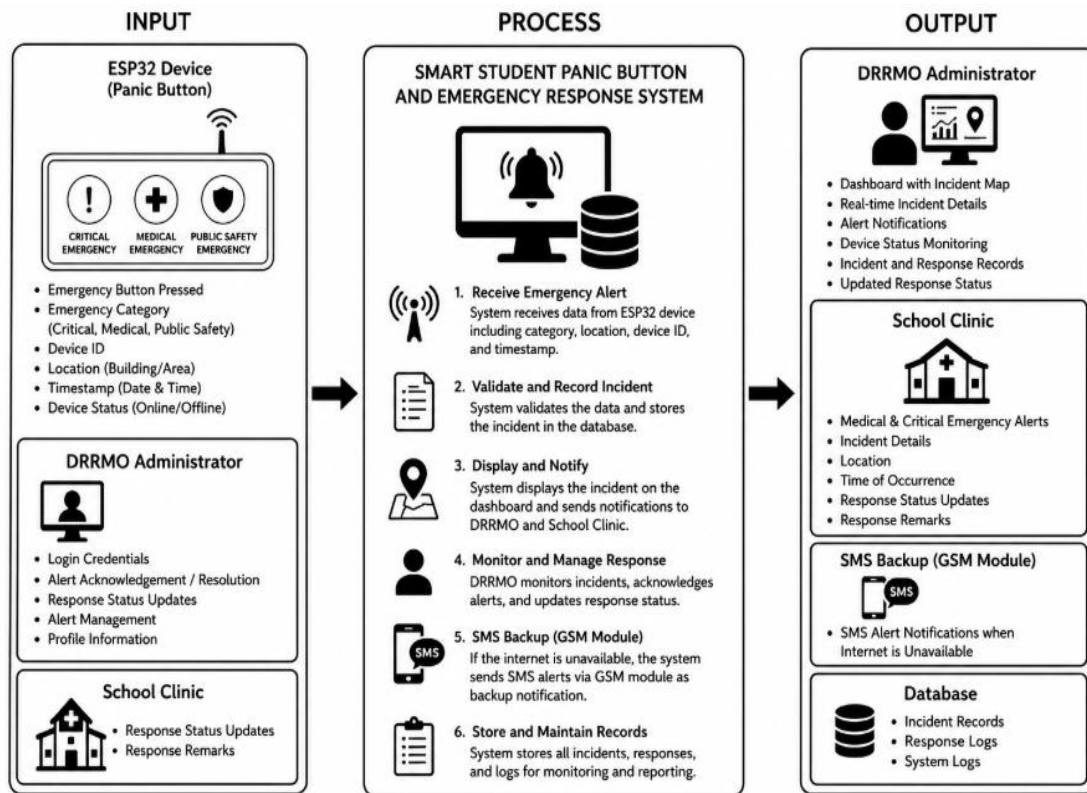


Figure 4. Conceptual Framework – IPO Diagram

The Input-Process-Output (IPO) Diagram of the Smart Student Panic Button and Emergency Response System with Campus Incident Mapping and SMS Backup Notification is shown in Figure 4. On the input, it includes data from the ESP32 panic button device, such as pressing of an emergency button, selected emergency category (Critical, Medical, or Public Safety), device ID, location, time, and device status. It also includes the input from the DRRMO Administrator and School Clinic. In the process stage, the system receives and validates the emergency data, logs the incident in the database, displays the incident on the web-based monitoring platform, notifies the appropriate responders, sends GSM-based SMS backup notifications when internet connectivity is unavailable and updates the emergency response status. The DRRMO Administrator receives a real-time monitoring dashboard, emergency notifications, details of incidents, campus incident map, device status, and response records, while the School Clinic receive medical and critical emergency notifications, incident information, location, and response

updates. The system also stores in the database the incident records, response logs and system logs for the purpose of monitoring and reporting.

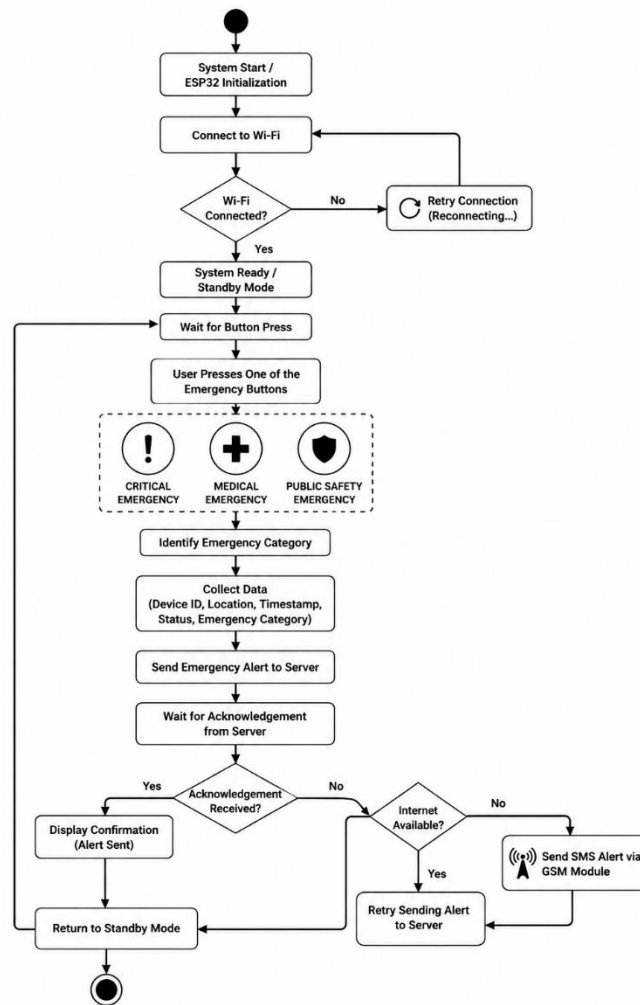


Figure 5. Activity Diagram (ESP32)

Figure 5. Activity flow of the ESP32 Panic Button Device in the proposed system. The process starts with the initialization of ESP32 and Wi-Fi connection. Once connected the device enters standby mode awaiting a user to touch one of the three dedicated emergency buttons; Critical Emergency, Medical Emergency or Public Safety Emergency. The selected emergency category is identified by the ESP32, collects the required information such as device ID, location, timestamp and device status and sends the emergency alert to the web-based monitoring platform. The device waits for an acknowledgement from the server. The alert is confirmed on receipt of an

acknowledgement and the device reverts to standby mode. In case of any internet connectivity problems, if the acknowledgement is not received, the ESP32 then triggers the GSM module to send an SMS backup notification before going to standby mode, providing reliable emergency reporting even in the case of network interruptions.

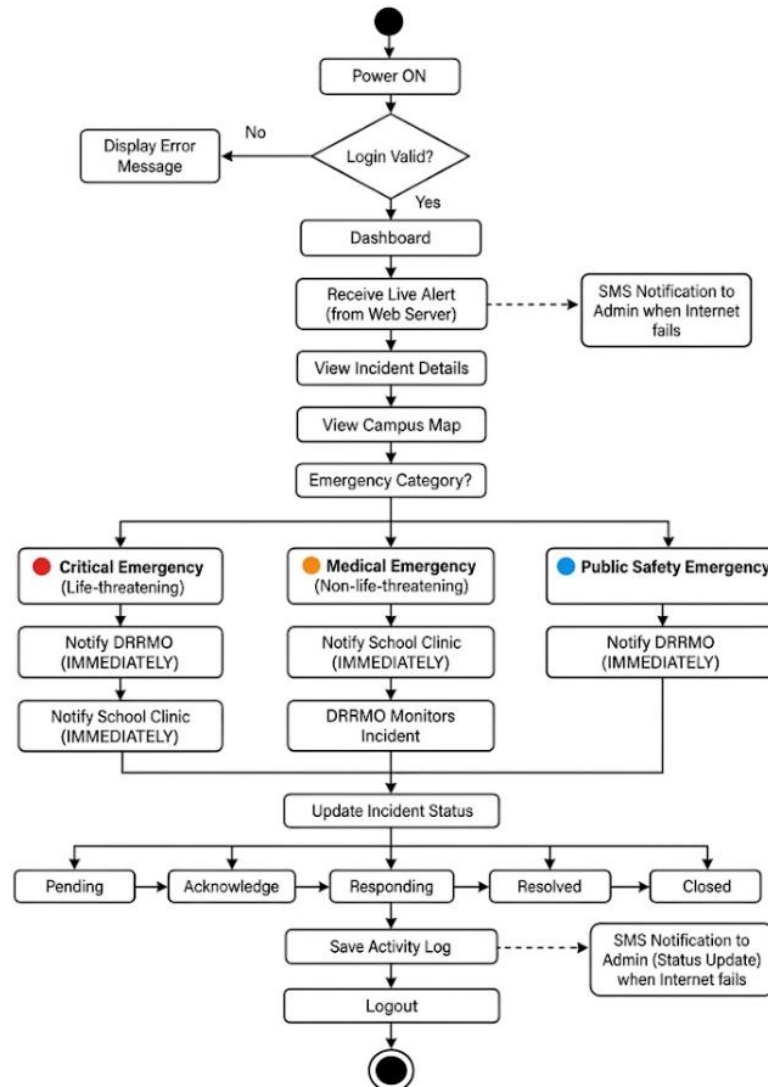


Figure 6. Activity Diagram (DRRMO Admin)

The Smart Student Panic Button and Emergency Response System with Campus Incident Mapping and SMS Backup Notification is developed by integrating embedded system and web design to create a web-enabled IoT-based system for emergency reporting and response management. The ESP32 microcontroller is mainly responsible for managing the panic button, receiving signals from three different panic buttons, processing the

wireless data, and sending the alerts to the remote web system. A SIM800L GSM module has been added to send SMS alerts in case of no network service. The firmware program written in C++ is installed on the ESP32 controller. It is written and developed with Arduino IDE. It controls the operation of the device functions, triggers event, sets the Wi-Fi communication with a remote server and starts the GSM connection. The program includes the WiFi.h library among others mainly to allow wireless communications with the monitoring server. The web front-end of the system is built with Native PHP, HTML, CSS and JavaScript for dynamic web functionality and MySQL has been chosen as the system database for all records. We use Laragon as the local web server and Visual Studio Code as the preferred IDE to run the development and modification of the web front-end. The web application enables DRRMO personnel to view the emergency requests received and respond to the incident locations using the campus incident map view.

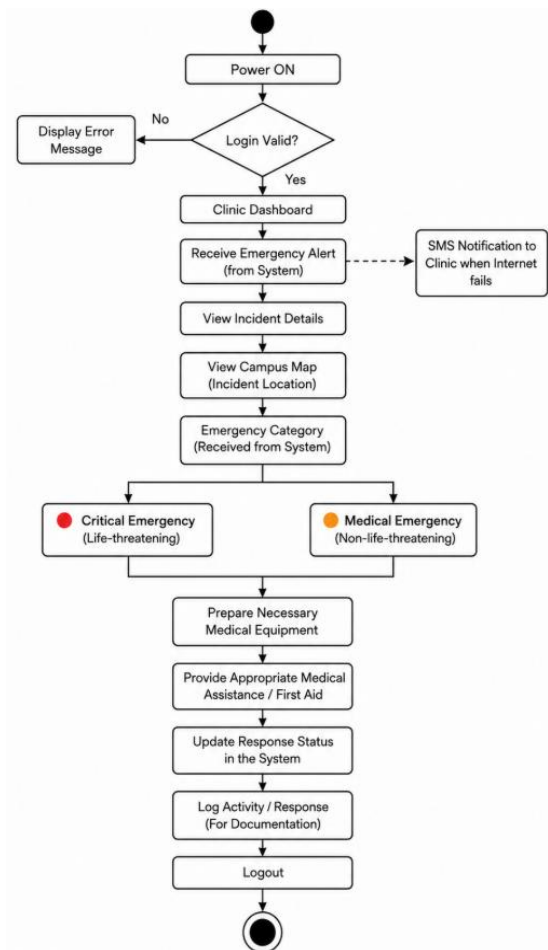


Figure 7. Activity Diagram (School Clinic)

Figure 7. Activity Diagram (School Clinic). Fig. 7 shows the step-by-step activities conducted by the School Clinic personnels in responding to Critical and Medical emergency incident in utilizing the Smart Student Panic Button and Emergency Response System. The clinic personnel will first log into the system. Invalid username/password will alert an error message whereas, a valid login will be able to see the clinic dashboard. Once the clinic personnel will receive a alert of emergency, the personnel will view the details of the incident and the location in the campus. The personnel then determines if the incident is classified as a Critical Emergency or Medical Emergency, prepares the necessary medical supplies, administer the first aid or the appropriate medical help. Once the response of the emergency is concluded, the personnel will update the response status of the incident, logs the activities for record-keeping, and finally log out from the system. The system will also auto send SMS notification back up in the absence of the internet, so the school clinic can receive alert information about an emergency.

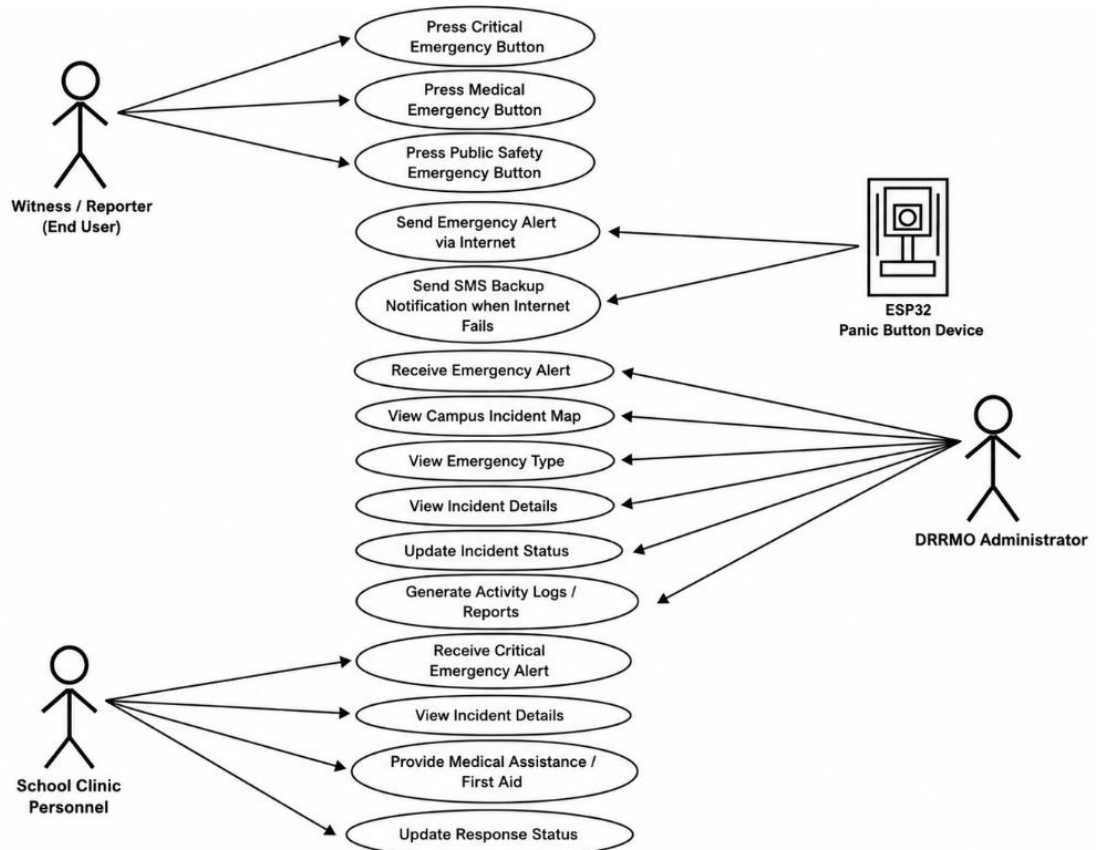


Figure 8. Use Case Diagram

Figure 8: Use Case Diagram Figure 8. Interactions of the actors with the Smart Student Panic Button and Emergency Response System. The Witness/Reporter starts an emergency by pressing one of the three dedicated emergency buttons for Critical Emergency, Medical Emergency, or Public Safety Emergency. Subsequently, the ESP32 Panic Button Device transmits the emergency alert over the internet and automatically sends an SMS backup notification when internet is not available. The DRRMO Administrator receives emergency alerts, views the campus incident map, emergency type, and incident details. They update the incident status and generate activity logs and reports to monitor and document emergency responses. Critical Emergency – The School Clinic Personnel receives a Critical Emergency alert, views the incident details, provides necessary medical assistance or first aid and updates the response status. Through these interactions, the system allows emergency reporting, coordinated response and efficient incident management within the J.H. Cerilles State College – Main Campus timely manner.

Table 3.3.5.1 Use Case Description – Press Critical Emergency Button

Name	Press Critical Emergency Button
ID	UC1
Description	The Witness/Reporter activates the Critical Emergency Button on the ESP32 Panic Button Device to report a life-threatening emergency requiring immediate response from both the DRRMO Administrator and the School Clinic Personnel.
Actor	Witness/Reporter (Students, Faculty, Staff, Visitors)
Preconditions	The ESP32 Panic Button Device is powered on, connected to the system, and ready for use.
Postconditions	A Critical Emergency alert is generated and transmitted to the DRRMO Administrator and the School Clinic Personnel through the internet or SMS backup when internet connectivity is unavailable.
Main Course	1. Press the Critical Emergency Button. 2. The ESP32 detects the button press. 3. The system retrieves the device ID, emergency type, and assigned location. 4. The system generates a Critical Emergency alert. 5. If the internet is available, the alert is transmitted to the DRRMO Administrator and the School Clinic Personnel through the web server. 6. The emergency incident is displayed on their dashboards for immediate response.
Alternate Course	AC1. Internet connection unavailable: 1. The system automatically switches to the GSM module. 2. An SMS backup notification is sent to the DRRMO Administrator and the School Clinic Personnel. AC2. ESP32 device failure: 1. The emergency alert cannot be generated. 2. The system waits until the device is restored.

Table 3.3.5.2 Use Case Description – Press Medical Emergency Button

Name	Press Medical Emergency Button
ID	UC2
Description	The Witness/Reporter activates the Medical Emergency Button on the ESP32 Panic Button Device to report a medical emergency that requires assistance from the School Clinic but is not immediately life-threatening.
Actor	Witness/Reporter (Students, Faculty, Staff, Visitors)
Preconditions	The ESP32 Panic Button Device is powered on, connected to the system, and ready for use.
Postconditions	A Medical Emergency alert is generated and transmitted to the DRRMO Administrator through the internet or SMS backup when internet connectivity is unavailable. The DRRMO Administrator coordinates the appropriate response.
Main Course	1. Press the Medical Emergency Button. 2. The ESP32 detects the button press. 3. The system retrieves the device ID, emergency type, and assigned location. 4. The system generates a Medical Emergency alert. 5. If the internet is available, the alert is transmitted to the DRRMO Administrator through the web server. 6. The emergency incident is displayed on the DRRMO dashboard for appropriate response.
Alternate Course	AC1. Internet connection unavailable: 1. The system automatically switches to the GSM module. 2. An SMS backup notification is sent to the DRRMO Administrator. AC2. ESP32 device failure: 1. The emergency alert cannot be generated. 2. The system waits until the device is restored.

Table 3.3.5.3 Use Case Description – Press Public Safety Emergency Button

Name	Press Public Safety Emergency Button
ID	UC3
Description	The Witness/Reporter activates the Public Safety Emergency Button on the ESP32 Panic Button Device to report an incident related to security, fire, accidents, or other public safety concerns that require immediate response from the DRRMO Administrator.
Actor	Witness/Reporter (Students, Faculty, Staff, Visitors)
Preconditions	The ESP32 Panic Button Device is powered on, connected to the system, and ready for use.
Postconditions	A Public Safety Emergency alert is generated and transmitted to the DRRMO Administrator through the internet or SMS backup when internet connectivity is unavailable.
Main Course	1. Press the Public Safety Emergency Button. 2. The ESP32 detects the button press. 3. The system retrieves the device ID, emergency type, and assigned location. 4. The system generates a Public Safety Emergency alert. 5. If the internet is available, the alert is transmitted to the DRRMO Administrator through the web server. 6. The emergency incident is displayed on the DRRMO dashboard for appropriate response.
Alternate Course	AC1. Internet connection unavailable: 1. The system automatically switches to the GSM module. 2. An SMS backup notification is sent to the DRRMO Administrator. AC2. ESP32 device failure: 1. The emergency alert cannot be generated. 2. The system waits until the device is restored.

Table 3.3.5.4 Use Case Description – Receive Emergency Alert

Name	Receive Emergency Alert
ID	UC4
Description	The DRRMO Administrator receives an emergency alert transmitted by the ESP32 Panic Button Device through the web server or SMS backup when internet connectivity is unavailable.
Actor	DRRMO Administrator
Preconditions	The DRRMO Administrator is logged in to the system and the monitoring platform is active.
Postconditions	The emergency alert is successfully received, displayed on the dashboard, and recorded in the system for appropriate action.
Main Course	1. The system receives the emergency alert from the ESP32 device. 2. The system validates the alert information. 3. The alert is displayed on the DRRMO dashboard. 4. The system records the incident in the database. 5. The DRRMO Administrator acknowledges the alert for appropriate action.
Alternate Course	AC1. Internet connection unavailable (SMS backup received): 1. The system receives the emergency alert via SMS backup. 2. The alert is manually encoded or automatically parsed (if supported). 3. The alert is displayed on the DRRMO dashboard. 4. The incident is recorded in the database. AC2. Invalid or incomplete alert data: 1. The system detects incomplete or invalid alert information. 2. The alert is marked as invalid and not displayed. 3. The system waits for a valid alert.

Table 3.3.5.5 Use Case Description – View Campus Incident Map

Name	View Campus Incident Map
ID	UC5
Description	The DRRMO Administrator views the campus incident map to identify the geographical location of the reported emergency based on the activated ESP32 Panic Button Device.
Actor	DRRMO Administrator
Preconditions	The DRRMO Administrator is logged in to the system and at least one emergency incident has been reported.
Postconditions	The location of the incident is displayed on the campus map to assist in faster response and decision-making.
Main Course	1. The DRRMO Administrator accesses the dashboard. 2. The system displays the campus incident map. 3. The system shows markers for all reported incidents. 4. The DRRMO Administrator selects an incident marker. 5. The system highlights the exact location of the incident on the map.
Alternate Course	AC1. No incident available: 1. The system displays an empty campus map. 2. The system shows a message “No incidents reported.” AC2. Map failed to load: 1. The system displays an error message. 2. The DRRMO Administrator refreshes the page. 3. The system reloads and displays the campus incident map.

Table 3.3.5.6 Use Case Description – View Incident Details

Name	View Incident Details
ID	UC6
Description	The DRRMO Administrator views detailed information of a selected incident including the emergency type, location, time, device information, and current response status.
Actor	DRRMO Administrator
Preconditions	The DRRMO Administrator is logged in to the system and at least one emergency incident has been reported.
Postconditions	The details of the selected incident are displayed for review and appropriate action.
Main Course	1. The DRRMO Administrator accesses the dashboard. 2. The system displays the list of reported incidents. 3. The DRRMO Administrator selects an incident from the list. 4. The system retrieves the details of the selected incident. 5. The system displays the incident details including emergency type, location, time, device ID, description, and response status. 6. The DRRMO Administrator reviews the information for appropriate action.
Alternate Course	AC1. Incident not found: 1. The system displays a message “Incident not found.” 2. The DRRMO Administrator returns to the incident list. AC2. Failed to load incident details: 1. The system displays an error message. 2. The DRRMO Administrator refreshes the page. 3. The system reloads and displays the incident details.

Table 3.3.5.7 Use Case Description – Update Incident Status

Name	Update Incident Status
ID	UC7
Description	The DRRMO Administrator updates the response status of an incident to reflect the progress of the emergency response until it is resolved or closed.
Actor	DRRMO Administrator
Preconditions	The DRRMO Administrator is logged in to the system and at least one emergency incident has been reported.
Postconditions	The incident status is successfully updated and recorded in the system. The updated status is reflected on the dashboard and incident records.
Main Course	1. The DRRMO Administrator accesses the incident details. 2. The system displays the current status of the incident. 3. The DRRMO Administrator selects a new status from the available options (e.g., Acknowledged, On-Going, Responding, Resolved, Closed). 4. The DRRMO Administrator enters additional remarks if needed. 5. The DRRMO Administrator saves the updated status. 6. The system records the update in the database. 7. The updated status is displayed on the dashboard and incident records.
Alternate Course	AC1. Invalid status selection: 1. The system detects an invalid status. 2. The system displays a message “Invalid status selected.” 3. The DRRMO Administrator selects a valid status. AC2. Update failed: 1. The system fails to save the updated status. 2. The system displays an error message. 3. The DRRMO Administrator retries the update.

Table 3.3.5.8 Use Case Description – Generate Activity Logs / Reports

Name	Generate Activity Logs / Reports
ID	UC8
Description	The DRRMO Administrator generates activity logs and reports of incidents and responses for monitoring, documentation, and future reference.
Actor	DRRMO Administrator
Preconditions	The DRRMO Administrator is logged in to the system and incident records exist in the database.
Postconditions	Activity logs and reports are successfully generated and can be viewed, downloaded, or printed.
Main Course	1. The DRRMO Administrator goes to the Reports/Logs section. 2. The system displays available report types (e.g., incident logs, response logs, summary reports). 3. The DRRMO Administrator selects the desired report type. 4. The DRRMO Administrator sets the date range or other filters if needed. 5. The DRRMO Administrator clicks the Generate button. 6. The system generates and displays the report with the requested data.
Alternate Course	AC1. No data available: 1. The system displays a message “No data available for the selected criteria.” 2. The DRRMO Administrator adjusts the filter or date range. AC2. Report generation failed: 1. The system displays an error message. 2. The DRRMO Administrator tries again to generate the report. 3. If the issue persists, the DRRMO Administrator contacts the system administrator.

Table 3.3.5.9 Use Case Description – Manage Users

Name	Manage Users
ID	UC9
Description	The DRRMO Administrator manages user accounts by adding, updating, deactivating, or deleting users who can access the Disaster Response Monitoring and Alert System.
Actor	DRRMO Administrator
Preconditions	The DRRMO Administrator is logged in to the system with sufficient privileges.
Postconditions	User accounts are successfully managed and the changes are recorded in the system.
Main Course	1. The DRRMO Administrator goes to the Users Management section. 2. The system displays the list of existing user accounts. 3. The DRRMO Administrator selects an action: Add, Update, Deactivate, or Delete. 4. The DRRMO Administrator enters or modifies the necessary user information. 5. The DRRMO Administrator saves the changes. 6. The system validates the input and updates the user records.
Alternate Course	AC1. Add User: 1. The DRRMO Administrator selects “Add User”. 2. The DRRMO Administrator enters the user details and role. 3. The system creates the new user account and notifies the user (if applicable). AC2. Update User: 1. The DRRMO Administrator selects a user. 2. The DRRMO Administrator updates the user information. 3. The system saves the updated information. AC3. Delete/Deactivate User: 1. The DRRMO Administrator selects a user to deactivate or delete. 2. The system marks the user as inactive or removes the account. 3. The user can no longer access the system (if deactivated/deleted). AC4. Invalid Input: 1. The system detects invalid or incomplete information. 2. The system displays an error message. 3. The DRRMO Administrator corrects the input and tries again.

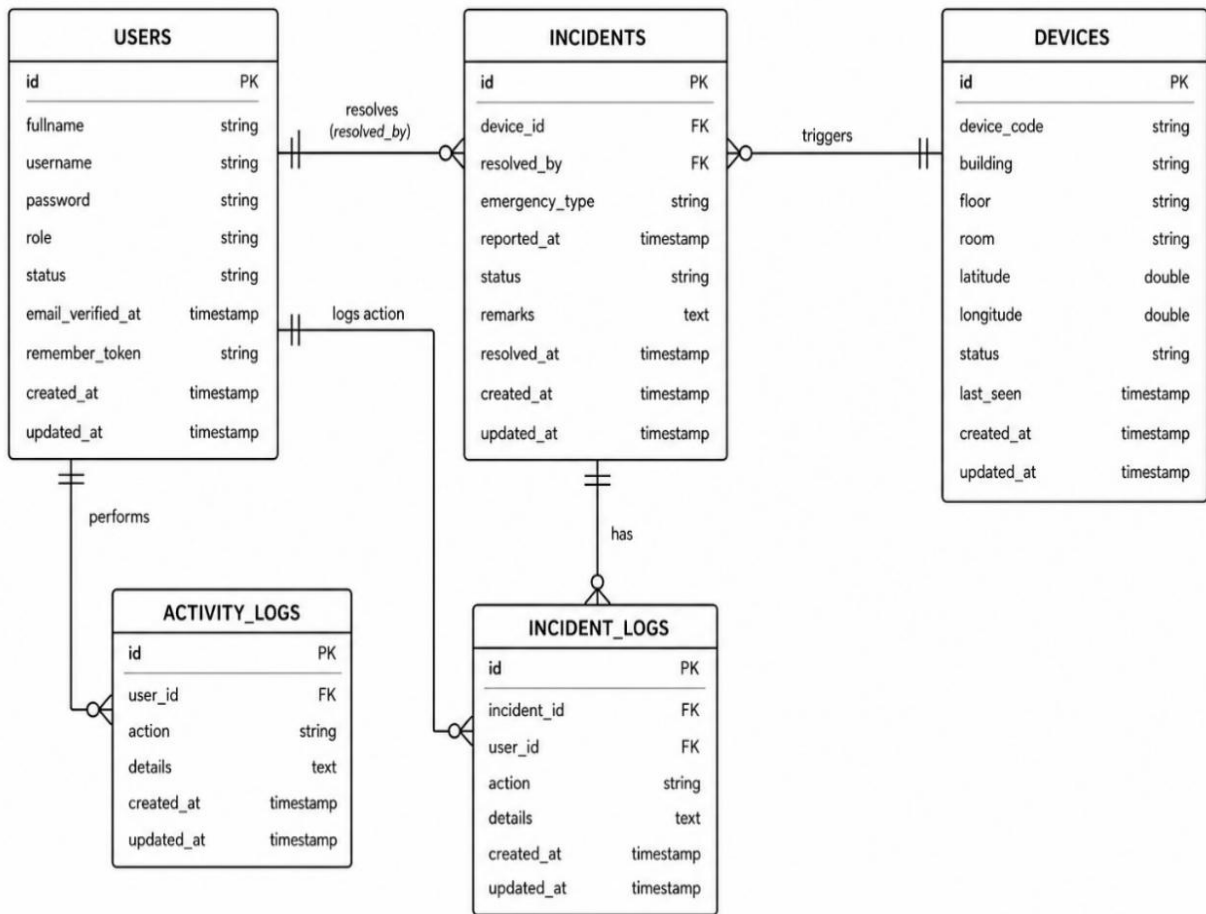


Figure 9. Entity Relationship Diagram

The database design for the proposed Smart Student Panic Button and Emergency Response system was developed following standard relational database modeling practices. Figure 10 demonstrates the database's structure, highlighting the inter-relationships between the system's five primary entities: Devices, Incidents, Users, Incident Logs and Activity Logs. The Devices entity houses the profile of each implemented ESP32 panic button, while Incidents serves to register every emergency that each device reports. DRRMO Administrator and School Clinic personnel's profile are found within the Users entity. Information regarding every step undertaken during emergency intervention are logged into the Incident Logs entity, and a record of all user interactions with the system is found in the Activity Logs entity. Each entity is connected with primary and foreign key constraints.

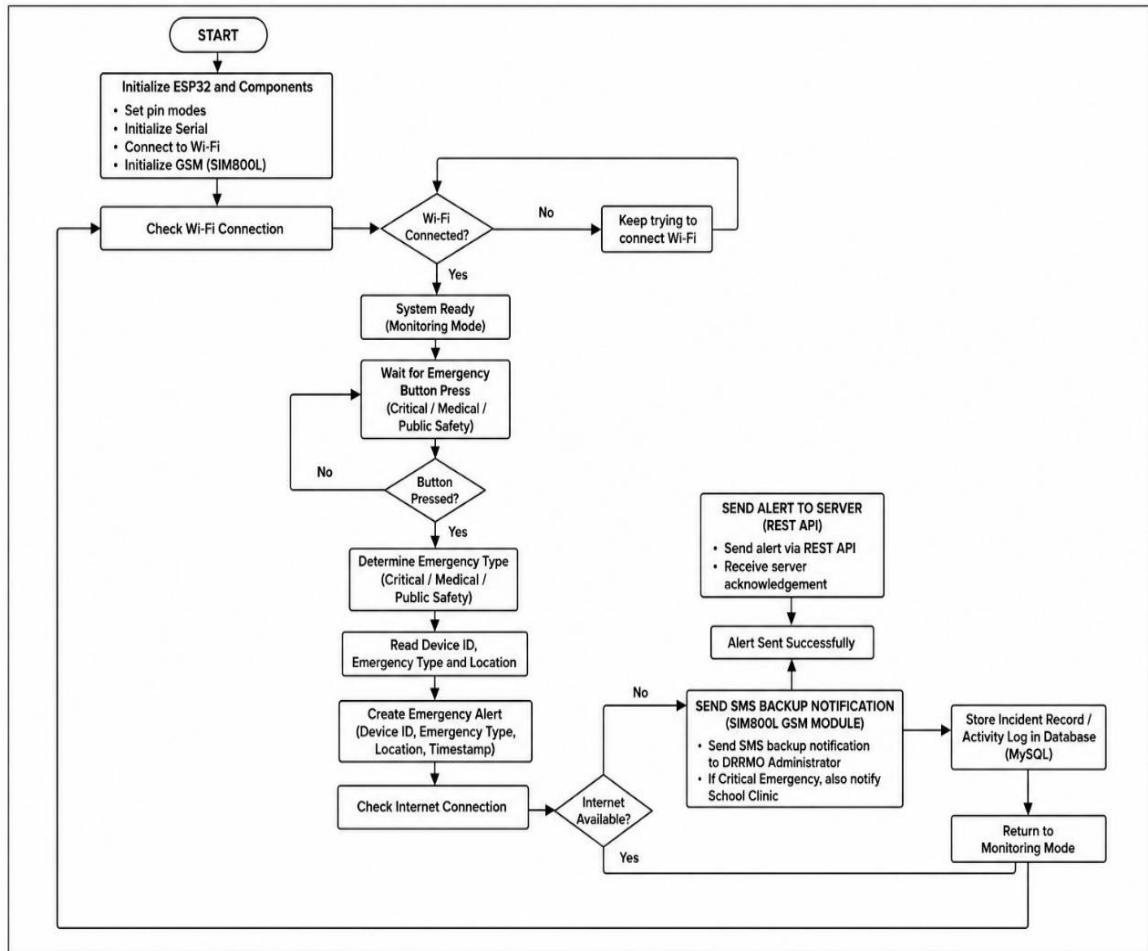


Figure 10. Structured Diagram

Figure 9. Organized Diagram. Figure 9 shows the process flow of the proposed Smart Student Panic Button and Emergency Response System. The process starts with the initialization of the ESP32, Wi-Fi connection and SIM800L GSM module. When the system switches to monitoring mode, it constantly listens for the pressing of one of the three emergency buttons: Critical Emergency, Medical Emergency and Public Safety Emergency. Upon pressing a button, the ESP32 identifies the type of emergency, retrieves the device ID and location, and generates an emergency alert. If an internet connection exists, the alert is sent to the web server via the REST API. Otherwise, the SIM800L GSM module will send an SMS backup notification to the DRRMO Administrator and in case of Critical Emergencies, to the School Clinic. Once the alert is successfully transmitted, the incident is stored in the database for monitoring and activity logging. The system then returns to monitoring mode, ready to respond to the next emergency.

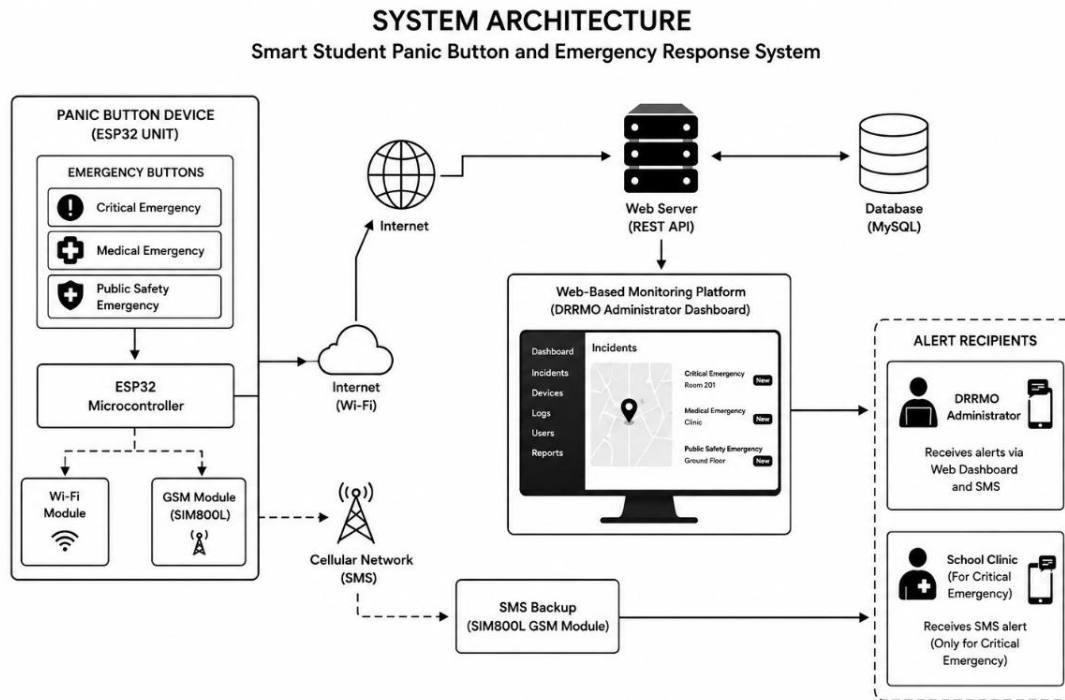


Figure 11. System Architecture

Figure 11 depicts the complete Architecture of the proposed Smart Student Panic Button and Emergency Response System. Firstly, the ESP32 Panic Button Device, an event reporter such as witness clicks one among the three (3) emergency buttons. Pressing either the “CRITICAL EMERGENCY”, “MEDICAL EMERGENCY” or “PUBLIC SAFETY EMERGENCY” buttons will initiate processing by the ESP32; thereafter sending via the internet the data to the web server through the REST API for storage on the MySQL Database, and then showing it to the DRRMO Administrator Dashboard for the incident to be monitored. In the event that there is no Internet access, an SMS alert would be triggered via the SIM800L GSM module as backup reporting of the emergency. The “CRITICAL EMERGENCY” notification will additionally be received by the School Clinic for their assistance.

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